

What is your counterparty's promise to pay worth?

CVA and DVA with simulation
and machine learning

Maurilio Patiño García

Sample class lecture notes

Universidad Anáhuac
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The demo examples are reproducible with the notebook that accompanies these notes;
the other figures come from the thesis and are illustrative.

Market data: Valmer (curves and CDS) and Bloomberg (swaptions).

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1 The problem: two identical swaps that are not worth the same

Picture two transactions that are exactly alike: a five-year interest rate swap, a notional of 10 million pesos, referencing TIIE de Fondo. The only difference is who signs on the other side. One is with a top-tier bank; the other, with a company whose credit the market considers fragile.

The contractual cash flows are identical. And yet the two transactions are **not worth the same**, because in the second one there is a higher probability that the counterparty will fail to pay what it owes when the time comes. That difference in value has a name: the **credit valuation adjustment**, or CVA.

The 2008 financial crisis put this adjustment at the centre of valuation. According to the Basel Committee (quoted in Gregory, 2012, sec. 17.4.4), roughly one third of counterparty risk losses in the crisis came from actual defaults; the other two thirds came from the **deterioration in the market value of the promise to pay** (Gregory himself notes that the empirical evidence for that proportion is not precise): counterparties that were still operating, but whose credit had deteriorated enough to reduce the value of their obligations sharply. That adjustment is recognised on the books even without a default.

CVA in practice

CVA is now part of fair value measurement (IFRS 13; in Mexico, NIF B-17 and C-10), Banco de México requires it monthly in its XVA report, and regulation assigns it capital (section 8). Many banks maintain teams dedicated to computing and hedging it, known as **CVA** or **XVA desks** (Gregory, 2012, ch. 18).

These notes develop the class in more depth than forty minutes allow. They assume familiarity with probability, present value and the basic mechanics of a swap, but not with stochastic calculus: every result that is needed is stated and explained, though not always proved.

2 The formula, in three pieces

The CVA of a transaction maturing at T is written

$$\text{CVA} = \underbrace{(1 - R)}_{\text{severity}} \int_0^T \underbrace{\text{DF}(t) \text{EE}(t)}_{\text{discounted exposure}} \underbrace{d\text{PD}(t)}_{\text{probability}} . \quad (1)$$

Here $\text{DF}(t)$ brings future losses to present value. The expression assumes exposure independent of default (no wrong-way risk). The swap is always valued in each scenario with the simulated rates; the approximation lies only in the aggregation. The exact form discounts each scenario with its own rates, $\mathbb{E}[D(0, t) V^+(t)]$; the expression uses the market discount factor times expected exposure, which omits the covariance between discounting and exposure. In the thesis, this approximation changes the reference swap's CVA by +0.25% (payer) and -0.35% (receiver).

The expression can be read as three common-sense questions, one per factor.

Severity: how much do I lose if it defaults? The factor $(1 - R)$ is the loss given default, where R is the recovery rate: the fraction of the amount owed that is recovered in bankruptcy proceedings. For emerging-market CDS, the convention is $R = 0.25$ (Iwashita, 2013): a quarter is recovered and three quarters are lost.

Exposure: how much will it owe me at each future date? The term $\text{EE}(t)$ is the expected exposure at time t : the average of what the counterparty would owe me if it defaulted

at that moment. This is the hard piece, because the future value of a swap depends on interest rates that do not yet exist. Section 4 is devoted to it.

Probability: how likely is it to default? The differential $dPD(t)$ is the probability that default occurs right around t . It is read off the credit derivatives market, as section 3 explains. In words: CVA is the average, across all future dates, of *what I would lose* multiplied by *the probability of losing it*. It can be seen as the price of implicit protection against the counterparty's default.

2.1 Why market probabilities and not historical ones

A natural question is why not use default probabilities estimated from historical data (observed default rates, rating transition matrices) instead of those implied by the credit market. The answer separates two activities that look alike but are not the same.

Forecasting answers how likely it is, in the real world, that this firm defaults. That is a statistical question, answered with historical data, under what the theory calls the physical measure $\mathbb{P}$.

Pricing answers how much it costs today to protect against that default. That is a price question, and the market supplies the answer: if an instrument exists that pays out when the counterparty defaults (a CDS) its quote contains all the information required. This is risk-neutral valuation, under the measure $\mathbb{Q}$.

The distinction people get wrong most often

CVA is a **price**, not a forecast: it is the discount applied to the value of a transaction, and it must be replicable by buying credit protection. That is why it is computed under $\mathbb{Q}$. CDS-implied probabilities are typically higher than historical ones because they embed the premium investors demand for bearing risk; that premium is part of the price and must not be stripped out.

The insurance analogy is best avoided. Insurance rests on risk mutualisation and is priced under $\mathbb{P}$; CVA rests on replication and is priced under $\mathbb{Q}$. Conflating them leads to wrong conclusions.

3 The probability is read off the market

A credit default swap (CDS) is a contract in which one party pays a periodic premium and, in exchange, receives compensation if a given issuer defaults. Its quote (the spread, in annual basis points) is literally the market price of that protection.

Definition 3.1 (Hazard rate). Default is modelled as the first jump of a counting process with intensity λ , called the hazard rate. Under that model, the probability of surviving to t is

$$Q(t) = e^{-\lambda t}. \quad (2)$$

The link to the market comes from equating, at inception, the present value of the CDS premiums with the present value of the expected compensation. Under a constant intensity that equality yields the approximate relation

$$\lambda \approx \frac{s}{1 - R}, \quad (3)$$

where s is the spread in decimal form.

Remark 3.2 (The intuition behind the ratio). If protection costs s per year and each default costs $(1 - R)$, then the implied default frequency is the ratio of the two. A spread of 150 basis points with a 25% recovery implies $\lambda = 0.015/0.75 = 2\%$ per year.

The probability of defaulting within an interval $[t_{i-1}, t_i]$ (what (1) denotes dPD) is simply the drop in the survival curve:

$$\Delta PD(t_i) = Q(t_{i-1}) - Q(t_i). \quad (4)$$

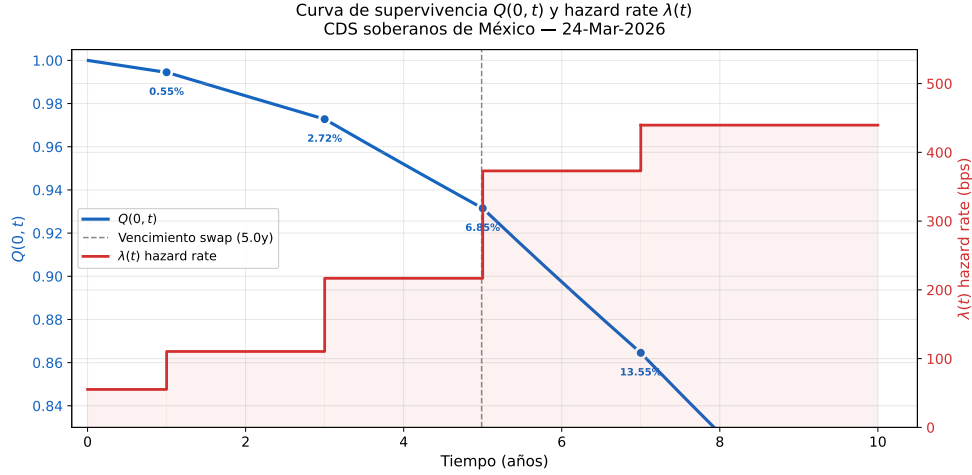


Figure 1: From the CDS spread to the survival curve. The bootstrap translates market quotes into default probabilities by tenor.

In practice the spread is not a single number: it is quoted at several tenors and the intensity is extracted piecewise, fitting each tenor sequentially so that all quotes are reproduced. That procedure is known as bootstrapping the credit curve. When the counterparty has no quoted CDS a proxy curve is used, chosen by the counterparty's rating, industry and region (Gregory, 2012). It is an approximation, and it is worth saying so.

4 Exposure: the future has to be simulated

To know how much the counterparty will owe me two years from now, I have to know what the swap will be worth two years from now, and that depends on the interest rates prevailing then. There is no way around it: scenarios of future rates have to be generated.

4.1 How many factors the curve needs

Before picking a model it pays to ask the data how many sources of uncertainty actually drive the curve. The standard tool is principal component analysis (PCA) applied to daily changes in rates across tenors.

Component	Variance explained	Interpretation
PC1	79.85%	Level: the whole curve moves up or down together
PC2	10.38%	Slope: short and long ends move in opposite directions
PC3	5.8%	Curvature: the belly moves relative to the wings

The reading is direct. In a single-factor model (Vasicek, Hull-White), every tenor depends on the same source of uncertainty. Tenors can move by different amounts (with mean reversion,

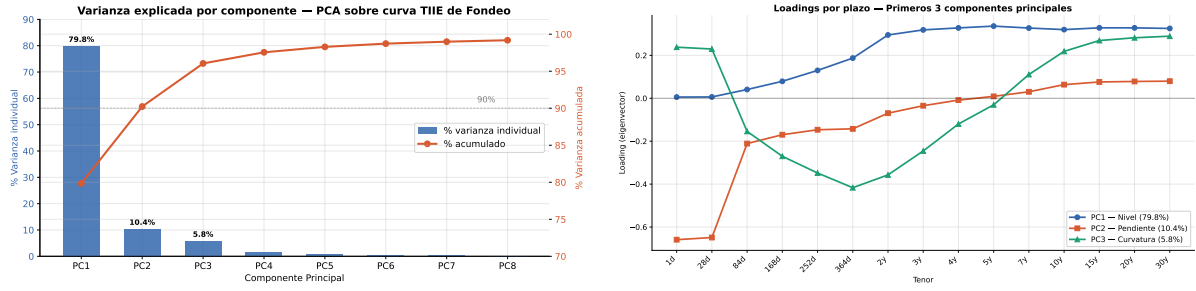


Figure 2: PCA of the TIIE de FONDEO curve. Left: variance explained by component. Right: loadings by tenor. The first component is roughly flat (level); the second changes sign between the short and the long end (slope).

the short end reacts more than the long end), but always in the same direction and with perfect correlation: the slope cannot move independently of the level.

Why it matters

For CVA this is not a cosmetic detail. The value of a swap depends on the shape of the curve, not just its level; a model that cannot deform the curve ignores a source of exposure variability. Level and slope together explain 90.23% of the variance: **two factors capture both movements**, and that is the working assumption.

4.2 The G2++ model

The G2++ model (Brigo and Mercurio, 2006) decomposes the instantaneous short rate into two correlated Gaussian factors plus a deterministic shift:

$$r(t) = x(t) + y(t) + \varphi(t), \quad (5)$$

with dynamics under the measure $\mathbb{Q}$

$$dx(t) = -a x(t) dt + \sigma dW_1(t), \quad x(0) = 0, \quad (6)$$

$$dy(t) = -b y(t) dt + \eta dW_2(t), \quad y(0) = 0, \quad (7)$$

$$dW_1(t) dW_2(t) = \rho dt. \quad (8)$$

Each factor is an Ornstein-Uhlenbeck process: it reverts to zero at speed a (or b) and receives random shocks of magnitude σ (or η). The shift $\varphi(t)$ is deterministic and is calibrated so that the model reproduces **exactly** the curve observed today; without it, the model would produce prices inconsistent with the market from day one.

The link between the PCA and the two factors

The mean reversion speed determines how each factor affects different tenors.

A slow factor (small mean reversion speed) persists over time, so its shocks affect rates at all tenors in a similar way: it is associated with the **level** (PC1).

A fast factor (large mean reversion speed) dies out before reaching the long tenors, so it moves the short end and leaves the long end still: it is associated with the **slope** (PC2).

The association is approximate: the G2++ factors are correlated, whereas the PCA components are orthogonal. Historical PCA motivates using two dimensions; it does not determine the dynamics under $\mathbb{Q}$.

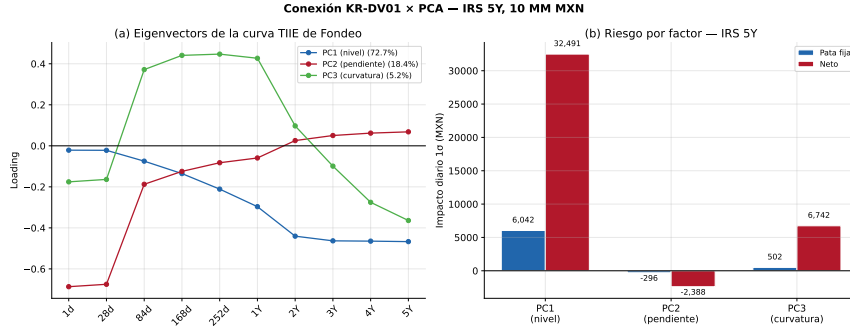


Figure 3: Link between the 5-year swap's KR-DV01 and PCA: (a) eigenvectors; (b) risk by factor. This PCA is estimated only on the ten nodes up to 5 years, where the swap has sensitivity; that is why its percentages (72.7%, 18.4% and 5.2%) differ from those of the full curve.

The parameters were calibrated to ATM swaptions with a reference date of 24 March 2026 (Bloomberg SWPM quotes, with USD SOFR discounting and MXN TIE-F RFR Curve projection), using the linear approximation of the swap rate variance. The model uses a single curve as a scope assumption. The parameters are:

Parameter	Value	Role
a	0.148849	mean reversion of the slow factor (level)
b	0.982642	mean reversion of the fast factor (slope)
σ	0.026973	volatility of the slow factor
η	0.062209	volatility of the fast factor
ρ	-0.913918	correlation between factors

Remark 4.1 (Reading the parameters). The fast factor reverts almost seven times more quickly than the slow one, which separates a level factor from a slope factor. The calibration used an unbounded Nelder-Mead search, so $\eta = 0.0622$ ended up outside the intended bound of 0.05.

4.3 Exact simulation per factor

A practical virtue of the G2++ is that its factors admit **exact simulation**, one at a time. Since each factor is Gaussian, its conditional distribution is known in closed form and it suffices to draw normals with the right mean and variance:

$$x(t + \Delta) = x(t) e^{-a\Delta} + \sqrt{\frac{\sigma^2}{2a} (1 - e^{-2a\Delta})} Z_1, \quad (9)$$

$$y(t + \Delta) = y(t) e^{-b\Delta} + \sqrt{\frac{\eta^2}{2b} (1 - e^{-2b\Delta})} Z_2, \quad (10)$$

where Z_1 and Z_2 are standard normals with correlation ρ , obtained from two independent normals via $Z_1 = \varepsilon_1$ and $Z_2 = \rho \varepsilon_1 + \sqrt{1 - \rho^2} \varepsilon_2$.

Remark 4.2 (Discretisation). Unlike an Euler-type scheme, here **each factor has no discretisation error, whatever the step size**. The correlation between innovations is not exact: using ρ instead of the exact correlation of the increments introduces a factor of 0.99996 on the thesis grid, a negligible effect. The relevant error is the Monte Carlo one, decaying as $O(1/\sqrt{N})$. This makes it possible to align the time grid with the swap's payment dates.

A contractual detail that moves the result

The floating-leg formula $1 - P(t, T_n)$ is exact only on payment dates: between two payments, the coupon already accrued is missing. With the grid aligned to the payments there is no bias; when the grid does not match the payments, the accrued coupon must be included. In the thesis portfolio, leaving it out understates book CVA by 10.3% (9.12 versus 10.17 MM MXN). That is why validating an engine matters.

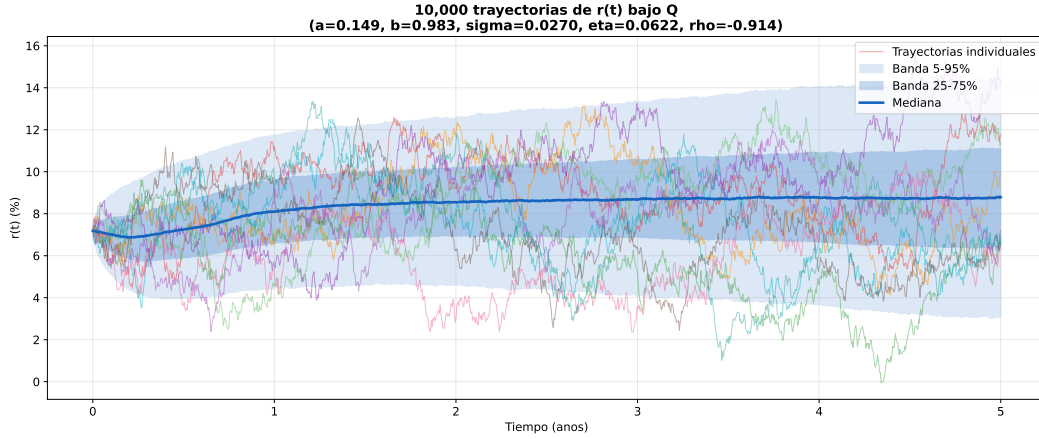


Figure 4: Ten thousand short rate paths under $\mathbb{Q}$ with the calibrated parameters. The fan widens with the square root of time: this is not a rate forecast, but the uncertainty the market is pricing today.

Once the factors are simulated, the price of any zero-coupon bond in any scenario follows in closed form:

$$P(t, T) = A(t, T) \exp(-B_1(t, T)x(t) - B_2(t, T)y(t)), \quad (11)$$

where $B_1(t, T) = (1 - e^{-a(T-t)})/a$ and likewise B_2 with b , while $A(t, T)$ collects the fit to the initial curve and a convexity correction. With this formula the full curve is rebuilt at every node of every scenario, and the swap is priced off it.

5 From scenarios to the hump

Pricing the swap in every scenario and at every date yields a distribution of values per date. Three measures are extracted from it:

$$EE(t) = \mathbb{E}[\max(V(t), 0)], \quad (12)$$

$$ENE(t) = \mathbb{E}[\max(-V(t), 0)]. \quad (13)$$

Expected exposure $EE(t)$ is computed as follows: in each scenario the value is kept if positive and replaced by zero if negative, and then *all* scenarios are averaged (it is not the mean of the positive values): if the swap is in my favour and the counterparty defaults, I lose that value; if it is against me, I lose nothing because I still owe. Hence the $\max(\cdot, 0)$. **Expected negative exposure** $ENE(t)$ is the mirror image. **Potential future exposure** (PFE) is a high percentile (in these notes, 97.5%) and is used to set credit lines, not to price.

5.1 The common origin of CVA and DVA

At any future date, the value of the swap across scenarios forms a distribution that, in the example of these notes, turns out roughly symmetric around zero (striking at the money does

not guarantee this in general). That single distribution, split at zero, generates both exposures.

Two risks, one simulation

The **positive side** is what the counterparty owes me. If it defaults, I lose. Its average is $EE(t)$, and it feeds the **CVA**.

The **negative side** is what I owe the counterparty. If I default, it loses. Its average is $ENE(t)$, and it feeds the **DVA**. In both cases the average is taken over all scenarios, with zeros where the side does not apply.

These are not two separate calculations and they do not require two models: they are two readings of the same simulation. For an at-the-money trade both exposures are nearly identical, so almost all the asymmetry in the final price comes from each party's credit spread.

Remark 5.1 (The non-linearity that complicates everything). The $\max(\cdot, 0)$ operator in (12) deserves a note, because almost every technical difficulty in this field is born there. Being non-linear, it makes CVA **non-additive**: the CVA of a netted portfolio is not the sum of the individual CVAs. That is why netting reduces exposure, and why computing the effect of adding a new trade to an existing portfolio is a problem in its own right.

5.2 Why exposure is hump-shaped

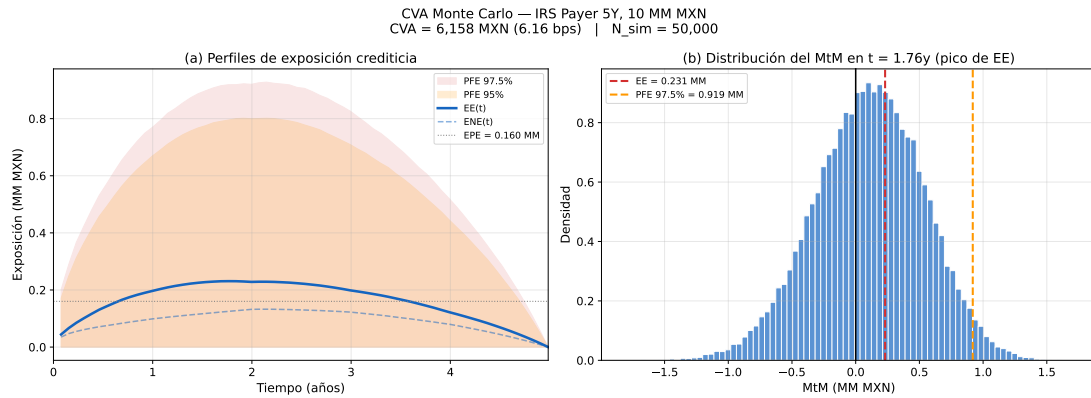


Figure 5: Exposure profiles of the reference swap. EE rises through rate diffusion and falls through amortisation; the peak appears around $T/3$. Illustrative figures.

The profile has a characteristic shape (it rises, peaks and decays to zero) that results from two opposing forces.

Diffusion pushes it up. The further out you look, the more uncertainty has accumulated about future rates, so the swap can drift further from zero in either direction. Dispersion grows roughly with $\sqrt{t}$.

Amortisation pushes it down. As time passes there are fewer cash flows left to exchange, so even if rates move a great deal, there is less to gain or lose. At maturity nothing remains: exposure is necessarily zero.

The maximum appears where the two forces balance, roughly at $T/3$ for a fixed-rate swap (Gregory, 2012, sec. 8.3.3). For the five-year reference swap, the peak occurs around $t = 1.8$ years (illustrative figure).

The operational conclusion

For the reference swap, **expected exposure does not peak at maturity**, but in the early part of the trade's life. The profile shows why exposure must be reviewed over the whole horizon, not only at the end.

6 The price: CVA, DVA and the bilateral adjustment

With both pieces in hand, CVA is computed by discretising (1) over the time grid:

$$\text{CVA} = (1 - R) \sum_i \text{EE}(t_i) \Delta \text{PD}(t_i) \text{DF}(t_i), \quad (14)$$

where $\text{DF}(t_i)$ is the market-curve discount factor (with the approximation described in the formula section). DVA is the same expression with the two substitutions already anticipated, the other profile and the other credit:

$$\text{DVA} = (1 - R) \sum_i \text{ENE}(t_i) \Delta \text{PD}^{\text{own}}(t_i) \text{DF}(t_i). \quad (15)$$

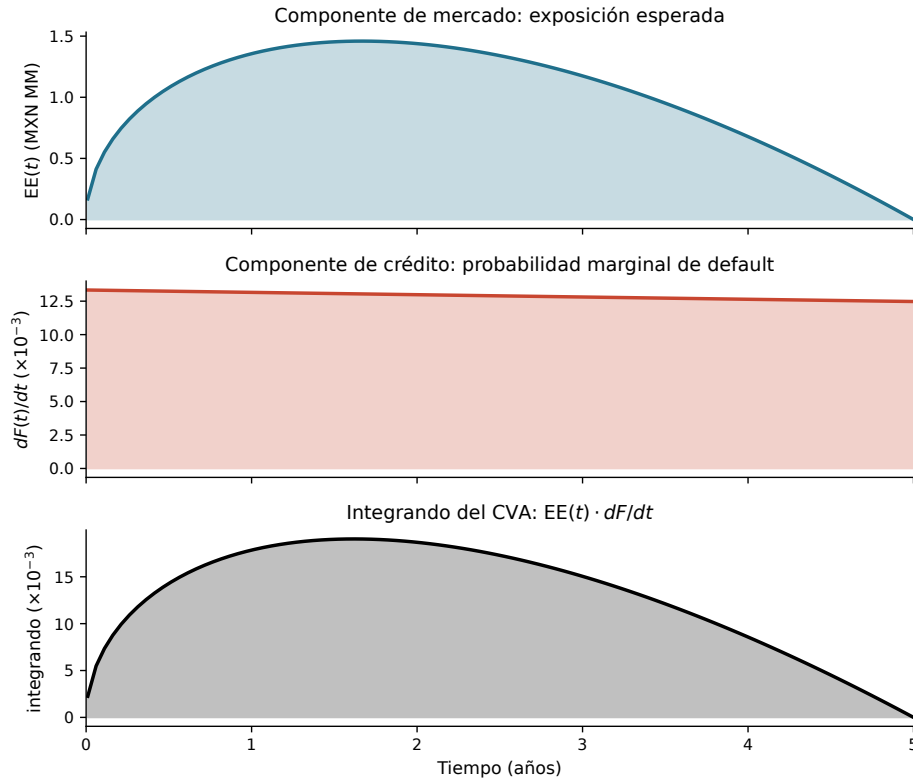


Figure 6: Anatomy of the integrand (conceptual sketch from the thesis, on a different scale from the reference swap): expected exposure (top), marginal default probability (middle) and their product (bottom). With discounting included, the area under the last curve, times $(1 - R)$, is the CVA.

The bilateral adjustment is the difference between the two:

$$\text{BCVA} = \text{DVA} - \text{CVA}. \quad (16)$$

This is a **simplified bilateral adjustment**: each term uses one party's marginal default probabilities and does not model which party defaults first, which is what determines the termination of the contract (Gregory, 2012).

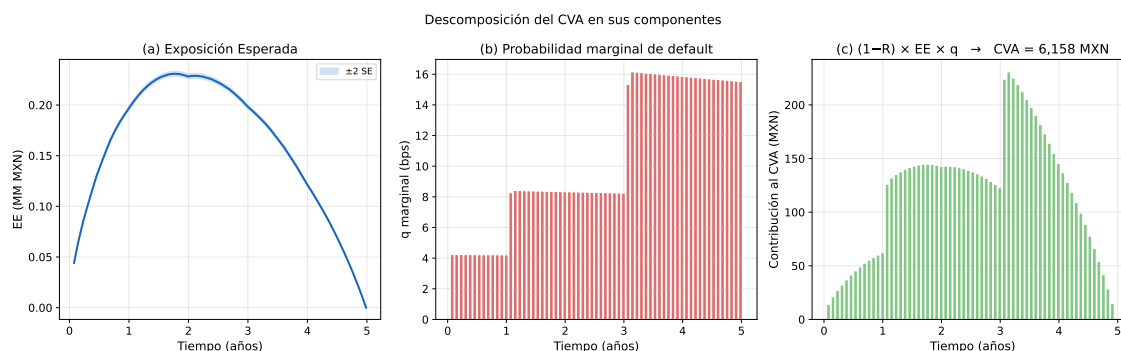


Figure 7: The reference swap: expected exposure, marginal default probability and each period's contribution, undiscounted (the 6,158 MXN CVA includes discounting). Axis labels in Spanish.

Example 6.1 (The demo swap). In the notebook demo (flat curve, illustrative figures), for the five-year payer swap on 10 million pesos, with the counterparty quoted at 150 basis points and ourselves at 80, the calculation gives $CVA = 7.65$ bps and $DVA = 4.21$ bps, hence $BCVA = -3.44$ bps: some 3,400 pesos coming straight out of the margin on the trade. The counterparty is the riskier of the two, so the net adjustment costs us.

Here is the answer to the question that always comes up in class: **yes, they combine into a single adjustment**, and the sign of the result depends on which of the two is riskier. If my counterparty trades wider than I do, CVA dominates and the adjustment costs me. If I trade wider than it does, DVA dominates and the adjustment works in my favour.

Remark 6.2 (Break-even point). If the two exposures are nearly symmetric and the recoveries are equal, the bilateral adjustment nearly vanishes when both spreads coincide. In the demo, the crossing occurs at about 148 bps, very close to the counterparty's 150 bps.

6.1 The awkwardness of DVA

DVA has a property worth stating plainly: **it is an accounting gain that increases as your own credit deteriorates**. If my institution weakens, my liabilities are worth less and that is booked as profit.

The reasoning is internally consistent (it reflects that my liabilities are worth less in the market), and IFRS 13 does include own credit risk when measuring liabilities (paragraph 42). But that gain is very hard to monetise (Gregory, 2012, sec. 13.3), which makes it useless as a solvency buffer. Along the same lines, Basel excludes own-credit adjustments from CET1 capital (CAP30.15), and regulatory CVA excludes the bank's own default (MAR50.3).

The thesis behind this class computes **unilateral CVA** as a *scope decision*, not because the standards require it: DVA appears here to complete the conceptual picture, not as a result of the work.

7 The operational problem and machine learning

With the engine described above, pricing the CVA of one swap takes on the order of tenths of a second. That sounds fast, until it is multiplied by the reality of a trading desk:

- A book of 795 trades requires minutes per full valuation.

- Computing sensitivities means repeating the calculation with the curve and the spreads bumped in dozens of directions: hours.
- A stress exercise with a thousand scenarios is measured in days of compute.

And it is not only about periodic reporting. The desk needs answers **during the day**: how much CVA is added by a trade being quoted right now, how to hedge the book, which counterparty the number comes from. At that scale, an engine taking hundreds of milliseconds per trade stops being viable.

7.1 Learning from the engine itself

The idea is to train a statistical model that learns the function mapping the features of a trade and of the market to its CVA. What makes the approach viable is where the training data comes from: **the Monte Carlo engine generates it**. There is no manual labelling and no external data; the teacher is the simulator.

Simulate once, price everything

The architecture is cheaper than it first appears. Since the cube of simulated paths can be reused to reprice the same swap under different credit spreads or different fixed rates, thousands of training examples cost a single simulation. That principle is what brings dataset generation down from hours to seconds.

Two model families are natural here. **Gradient boosted trees** (XGBoost) are hundreds of decision trees in which each one corrects the errors of the previous: widely used with tabular data. **Neural networks** are compositions of linear transformations and non-linear activations; they produce a continuous function (piecewise linear with ReLU activations), which, as we shall see, matters.

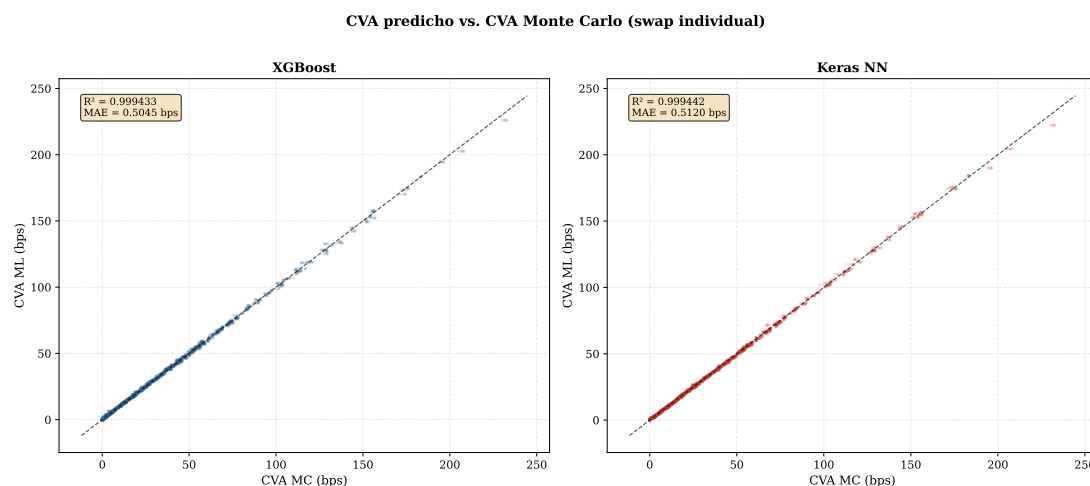


Figure 8: Predicted CVA against Monte Carlo CVA for both models. Each point is a swap from the test set; the dashed line is the identity.

Method	Time per swap (batch)	Speed-up	R^2
Monte Carlo	tenths of a second	$1\times$	benchmark
Analytical approximation	same order	a few times	n/a
XGBoost	microseconds	10^3 to $10^5\times$	> 0.999
Neural network	microseconds	10^3 to $10^5\times$	> 0.999

Thesis wall-clock times, on a single machine and in its configuration: average per query when pricing in batch.
Training is excluded.

7.2 Outside the training grid

Inside the training grid, XGBoost and the neural network tie (MAE of 0.50 and 0.55 bps). Outside it, against labels computed with 50,000 paths, the gap is large: with every variable outside the grid, XGBoost's MAE rises to 4.43 bps while the network's stays at 0.66 bps. XGBoost is piecewise constant and saw only ten tenors, so its prediction does not change between two trained tenors. The network's continuity is an explanation consistent with its better result, although on its own it does not guarantee better generalisation.

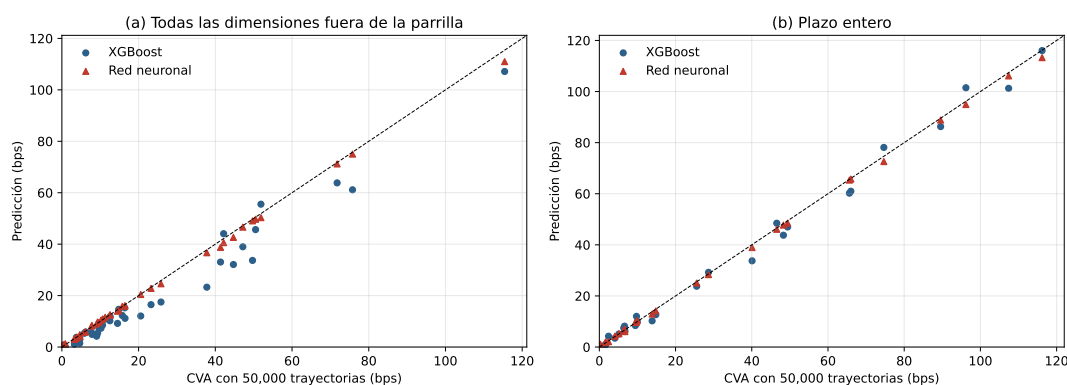


Figure 9: Prediction versus CVA with 50,000 paths on configurations outside the training grid: (a) all dimensions outside; (b) integer tenor.

7.3 How to read the speed-up

A speed-up of several orders of magnitude demands precision about what it measures, because it is easy to oversell. In the thesis it is the **average cost per query when pricing in batch**, with a model already trained and resident in memory. An isolated query takes longer, because of each call's fixed overhead. And the comparison depends on the reference engine: if the cube of paths can be reused (because only the spread or the fixed rate changes), the engine also answers in milliseconds; the surrogate pays off when a query forces a new simulation. It excludes generating the labels, training, validating and periodic retraining.

Over the full lifecycle, the break-even point is the preparation cost divided by the saving per query (in the notebook demo, a few tens or hundreds of queries). For an isolated query, by contrast, the Monte Carlo engine remains cheaper end to end. The speed-up applies **in the repeated-use regime**.

The role of the engine

The engine generates the labels, validates the surrogate and remains the reference for reporting. The surrogate makes it possible to evaluate many scenarios in little time.

7.4 A result worth knowing: the model depends on the question

Trees predict the *level* of CVA with remarkable accuracy. However, if one tries to obtain a *sensitivity* (the derivative of CVA with respect to the curve or the spread) from that same model trained on levels, the result is unreliable: in the thesis, XGBoost errors are 33.2% in

CS01, 76.8% in the PC1 PV01 and 545.7% in the PC2 PV01, the latter with the wrong sign.

Remark 7.1 (The reason is structural). A tree is a step function: constant within each leaf, with jumps at the boundaries. Its derivative is zero almost everywhere and undefined at the jumps. That fingerprint is visible to the naked eye in a CVA heat map generated with trees: the contour lines are staircases, not smooth curves.

The conclusion is not that trees are bad, but that **the right model depends on what you will ask of it**. For levels, trees are excellent. For sensitivities, a model whose derivative exists and is stable is preferable. The thesis uses a local Ridge regression, with errors median errors of 0.15% to 2.31%, but only within its training domain (shocks of up to $\pm 0.3\sigma$ per factor); larger shocks are extrapolation.

8 The Mexican regulatory framework

CVA is not only an academic exercise. In Mexico, these references are worth distinguishing.

IFRS 13 (fair value). Fair value reflects non-performance risk, which includes own credit risk when measuring liabilities (paragraph 42), and the standard addresses the net exposure to a counterparty (paragraph 56). That is why unilateral CVA is a scope decision, not a requirement of the standard.

NIF B-17 and C-10 (Annex 33 of the CUB, CNBV). For credit institutions, NIF B-17 (fair value measurement) and C-10 (derivative financial instruments) apply through criterion A-2, paragraph 2, of Annex 33.

Banco de México's XVA report. According to its instructions (March 2022), it is filed monthly, by counterparty or by trade type, with every valuation adjustment the institution computes. Its legal basis is article 36 of the Banco de México Act and article 97 of the Credit Institutions Act. It does not ask for sensitivities or scenario recalculations.

Banco de México Circular 4/2012, Annex 1. It requires monitoring the risk of credit valuation adjustments (requirement 2) and real-time risk control that includes that adjustment, so that the impact of new trades is known (requirement 11). This is the basis for pre-deal incremental CVA.

CVA capital. In Mexico it is set by articles 2 Bis 98 a to 2 Bis 98 d of the CUB, with a formula based on risk weights, effective maturity and discounted exposure; it does not use sensitivities. The Basel framework (MAR 50) is the international reference.

Two different things that share a name

Accounting CVA must not be confused with regulatory capital for CVA risk. The first is a valuation adjustment under the measure $\mathbb{Q}$, reported in Banxico's form. The second is a capital requirement. Under Basel, SA-CVA uses sensitivities of *regulatory* CVA, not accounting CVA (MAR50.29), and regulatory CVA excludes the bank's own default (MAR50.3).

One piece of local context: the instrument in this class references THIE de Fondeo, the one business day rate published by Banco de México (series SF331451). CME has cleared OIS on this rate since 5 February 2024 (CME Group, *Conversion Pricing for Cleared MXN 28D THIE Swaps*, March 2024, p. 3), and MexDer and Asigna announced the launch of the THIE de Fondeo swap on 26 February 2024 (press release of that date).

9 Summary

1. Two transactions with identical cash flows are not worth the same if the counterparties differ in credit quality. That difference is the CVA.
2. CVA decomposes into severity, exposure and probability. The probability is read off the CDS market because CVA is a price, not a forecast.
3. Exposure requires simulating the future. PCA of the THIE de Fondeo curve shows that two factors (level and slope) explain 90.23% of the variance, which motivates the G2++.
4. Expected exposure is hump-shaped; for the reference swap it peaks in the early part of the trade's life, not at maturity.
5. CVA and DVA are two readings of the same simulated distribution, split at zero. They combine in the bilateral adjustment, whose sign depends on which party is riskier.
6. At portfolio scale and with intraday needs, the Monte Carlo engine is not enough. A surrogate trained on labels from the engine itself answers much faster, especially in batch; the engine remains the reference for validation and reporting.
7. The right model depends on the question: trees for levels, local Ridge for sensitivities within its domain.

10 Further reading

For those who want to go deeper, these references are ordered from most accessible to most technical.

- Gregory, J. (2012). *Counterparty Credit Risk and Credit Value Adjustment*. Wiley. A reference text on the topic; readable without advanced stochastic calculus.
- Green, A. (2015). *XVA: Credit, Funding and Capital Valuation Adjustments*. Wiley. Covers the full family of valuation adjustments.
- Brigo, D. and Mercurio, F. (2006). *Interest Rate Models: Theory and Practice*. Springer. The source of the G2++ and of the closed-form results used here.
- Litterman, R. and Scheinkman, J. (1991). Common factors affecting bond returns. *The Journal of Fixed Income*. A classic paper on the level, slope and curvature interpretation.